Task S5.01. Queries with MongoDB

Description

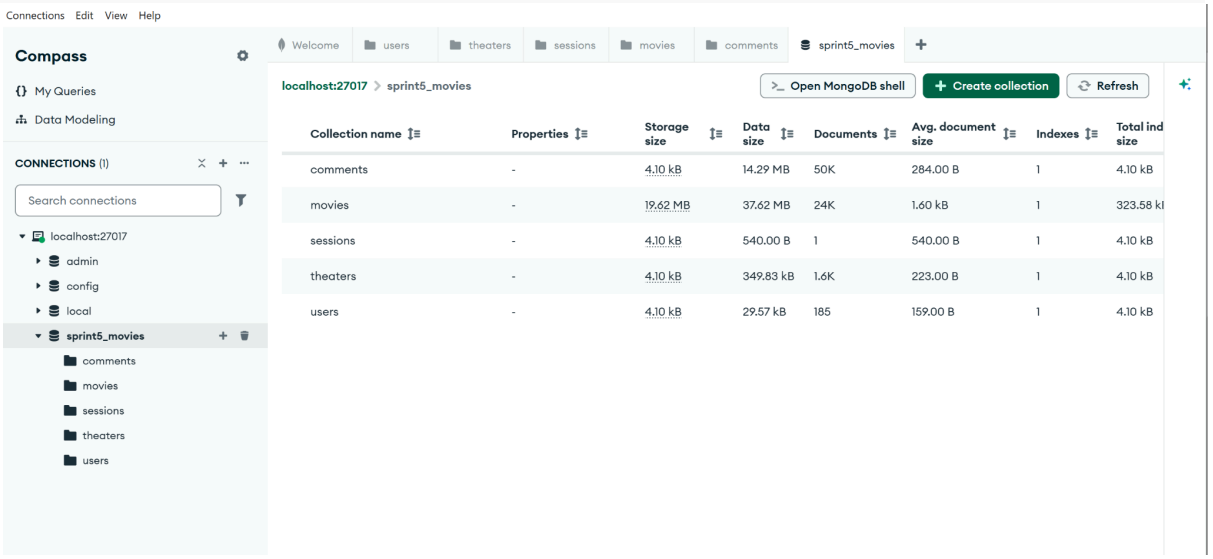
We will work with a database that contains collections related to a film entertainment application:

- users: Stores user information, including names, emails, and encrypted passwords.
- theatres: Contains cinema data, such as ID, location (address and geographical coordinates).
- sessions: Stores user sessions, including user IDs and JWT tokens for authentication.
- movies: Includes movie details, such as plot, genres, duration, cast, comments, release year, directors, ratings and awards.
- comments: Stores user comments about movies, with information about the author of the comment, movie ID, comment text, and date.

You will carry out some queries requested by the client, who is measuring whether or not you will be able to take charge of the analytical part of the project linked to their database.

Level 1

Create a database with MongoDB using the attached files as collections.



Exercise 1

- Shows the first 2 comments in the database.

```
>_MONGOSH
> use("sprint5_movies")
< switched to db sprint5_movies
> db["comments"].find().limit(2)
< {
    _id: ObjectId('5a9427648b0beebeb69579cc'),
    name: 'Andrea Le',
    email: 'andrea_le@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabcd418c'),
    text: 'Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptat
    date: ISODate('2012-03-26T23:20:16.000Z')
  }
  {
    _id: ObjectId('5a9427648b0beebeb69579cf'),
    name: 'Greg Powell',
    email: 'greg_powell@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabcd41b1'),
    text: 'Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate n
    date: ISODate('1987-02-10T00:29:36.000Z')
  }
sprint5_movies>
```

Documents 50K | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query | Explain | Reset | Find | Options

Project { field: 0 }

Sort { field: -1 } or [['field', -1]] | Max Time MS 60000

Collation { locale: 'simple' } | Skip 0 | Limit 2

Index Hint { field: -1 } or "indexName"

ADD DATA | BULK | EXPORT DATA | EXPORT CODE | 25 | 1 – 2 of 2

```
_id: ObjectId('5a9427648b0beebeb69579cc')
name: "Andrea Le"
email: "andrea_le@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd418c')
text: "Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate und…
date: ISODate('2012-03-26T23:20:16.000+00:00') March 26, 2012 at 11:20:16 PM UTC
```

```
_id: ObjectId('5a9427648b0beebeb69579cf')
name: "Greg Powell"
email: "greg_powell@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd41b1')
text: "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate neque.…
date: ISODate('1987-02-10T00:29:36.000+00:00') February 10, 1987 at 12:29:36 AM UTC
```

localhost:27017 > sprint5_movies > comments | Open MongoDB shell

Documents 50K | Aggregations | Schema | Indexes 1 | Validation

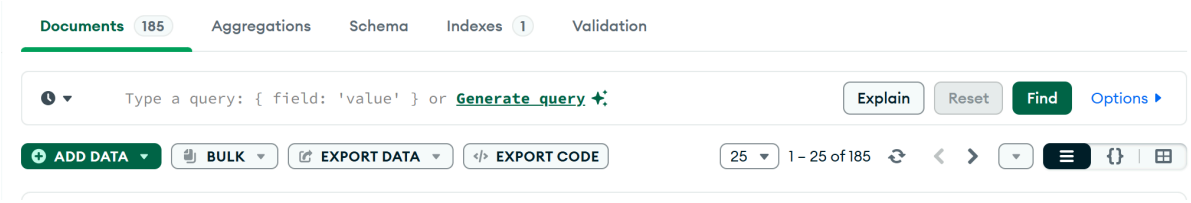
$limit | Edit | Explain | Run | Options

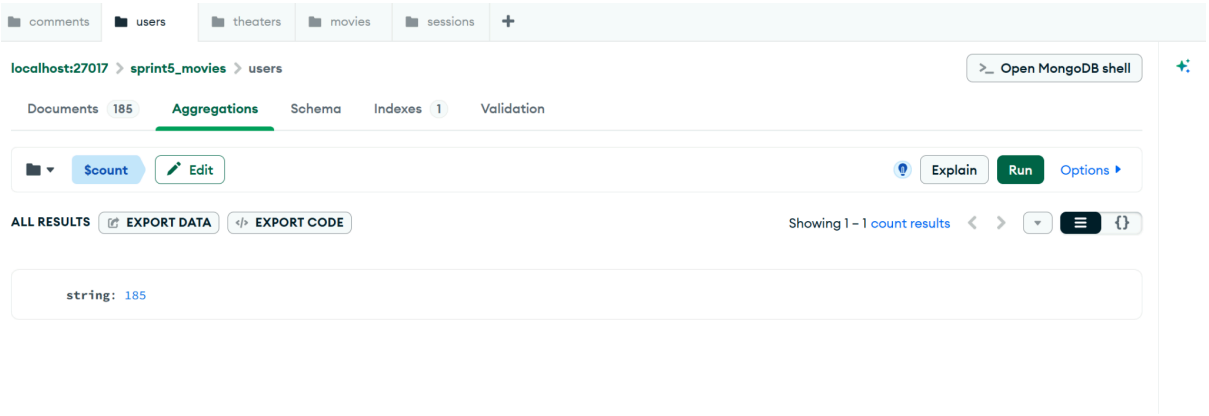
ALL RESULTS | EXPORT DATA | EXPORT CODE | Showing 1 – 2 count results

```
_id: ObjectId('5a9427648b0beebeb69579cc')
name: "Andrea Le"
email: "andrea_le@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd418c')
text: "Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate unde nulla temporibus…
date: ISODate('2012-03-26T23:20:16.000+00:00') March 26, 2012 at 11:20:16 PM UTC
```

```
_id: ObjectId('5a9427648b0beebeb69579cf')
name: "Greg Powell"
email: "greg_powell@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd41b1')
text: "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate neque. Dolorum nihil ver…
date: ISODate('1987-02-10T00:29:36.000+00:00') February 10, 1987 at 12:29:36 AM UTC
```

- How many users do we have registered? **185 users**

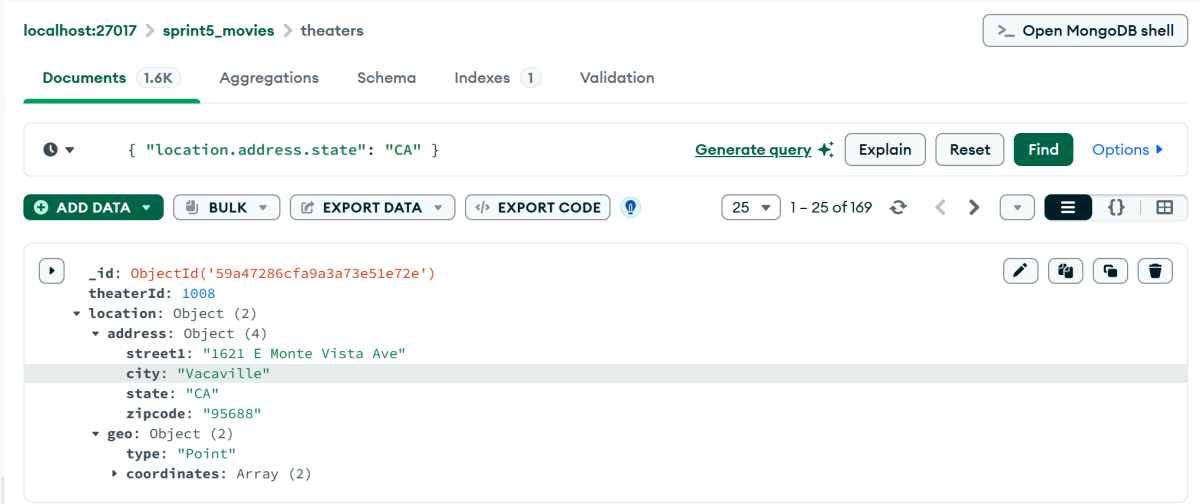


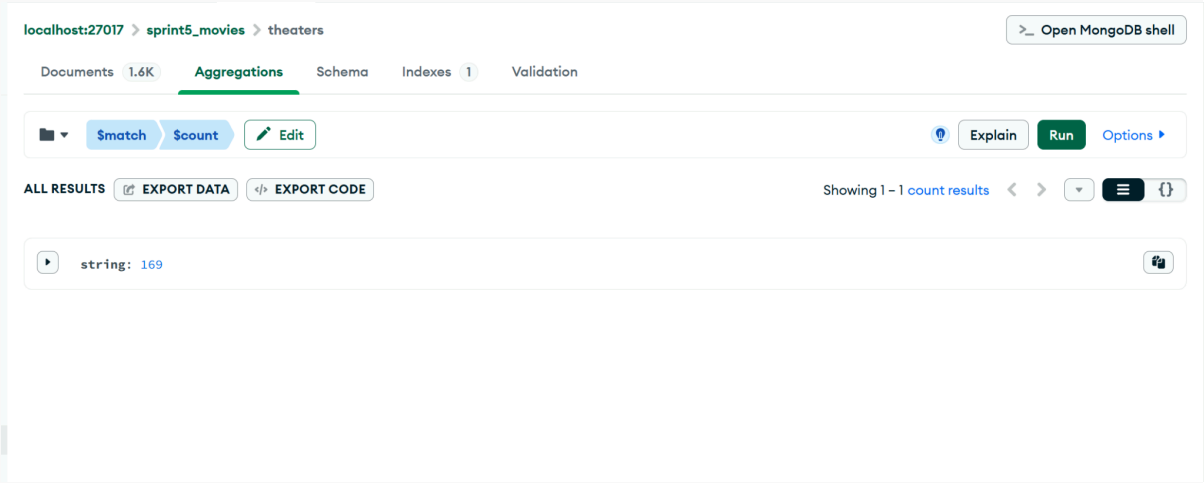




- How many movie theaters are there in the state of California? **169**

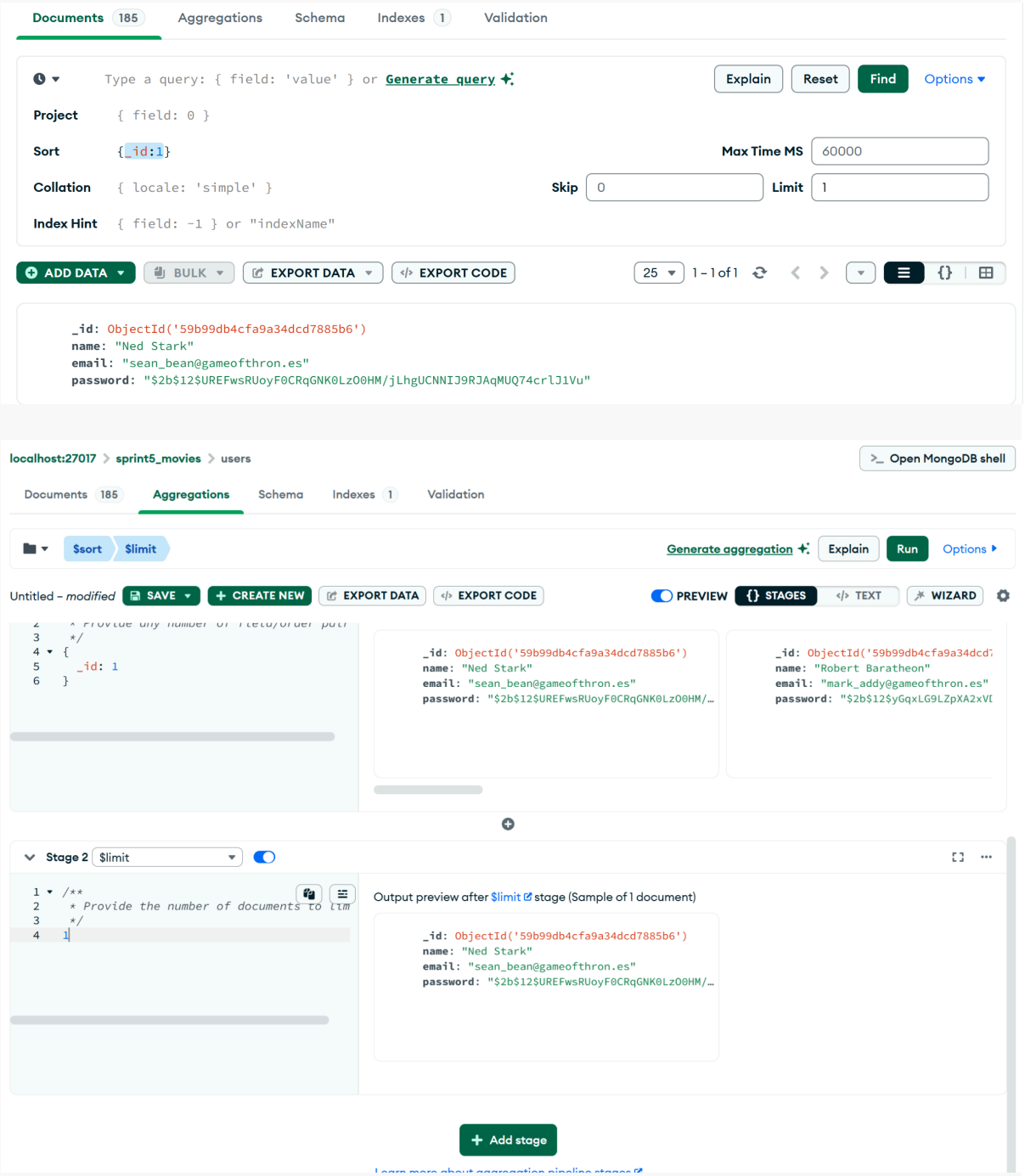
```
> db.theaters.distinct("location.address.state")
< [
    'AK', 'AL', 'AR', 'AZ', 'CA', 'CO', 'CT',
    'DC', 'DE', 'FL', 'GA', 'HI', 'IA', 'ID',
    'IL', 'IN', 'KS', 'KY', 'LA', 'MA', 'MD',
    'ME', 'MI', 'MN', 'MO', 'MS', 'MT', 'NC',
    'ND', 'NE', 'NH', 'NJ', 'NM', 'NV', 'NY',
    'OH', 'OK', 'OR', 'PA', 'PR', 'RI', 'SC',
    'SD', 'TN', 'TX', 'UT', 'VA', 'VT', 'WA',
    'WI', 'WV', 'WY'
  ]
> db.te
< sprint5_movies.te
> db.theaters.countDocuments({
    "location.address.state":"CA"
  })
< 169
```





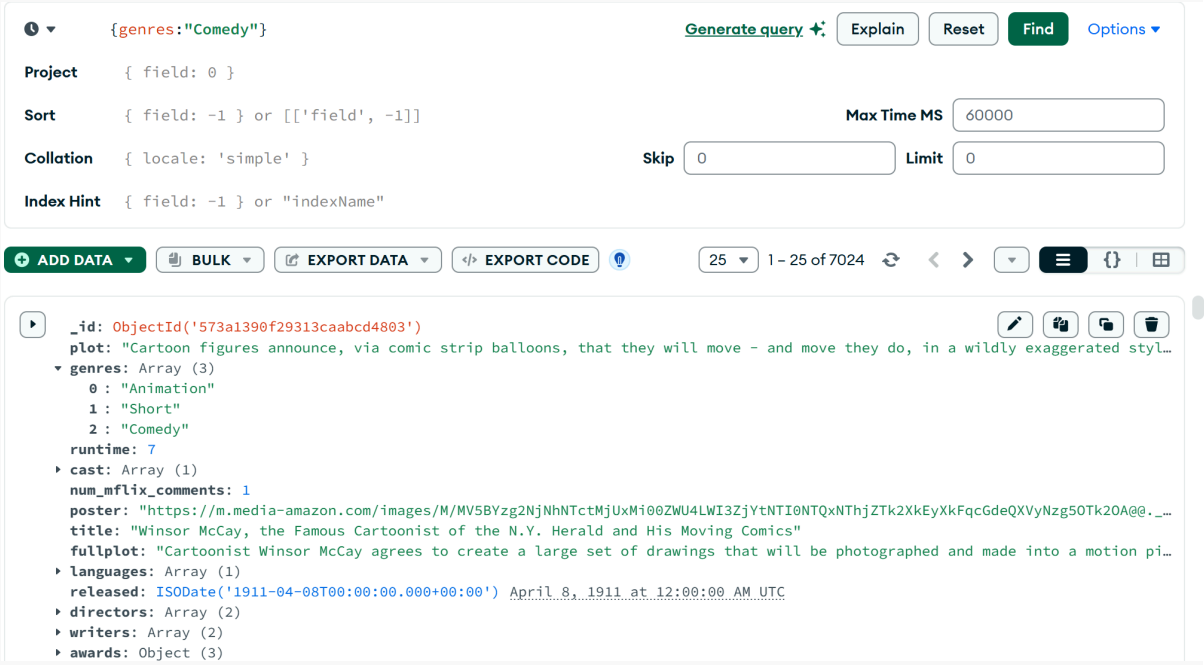
- Who was the first user to register? **Ned Stark**

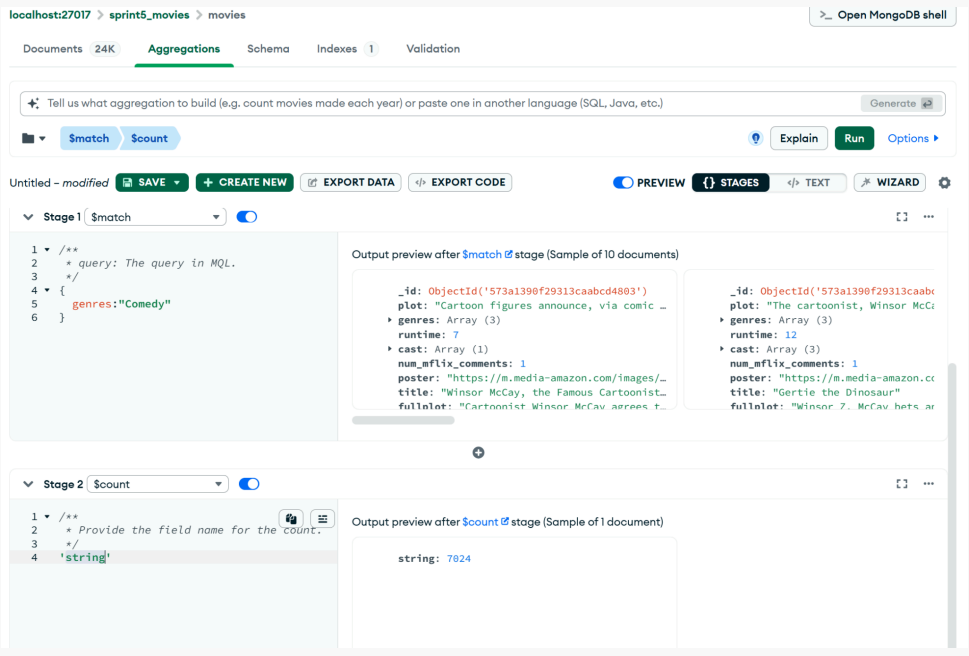
```
> db.users.find().sort({
    _id:1
  }).limit(1)
< {
    _id: ObjectId('59b99db4cfa9a34dcd7885b6'),
    name: 'Ned Stark',
    email: 'sean_bean@gameofthron.es',
    password: '$2b$12$UREFwsRUoyF0CRqGNK0LzO0HM/jLhgUCNNIJ9RJAqMUQ74crlJ1Vu'
  }
```



- How many comedy films are in our database? **7024**

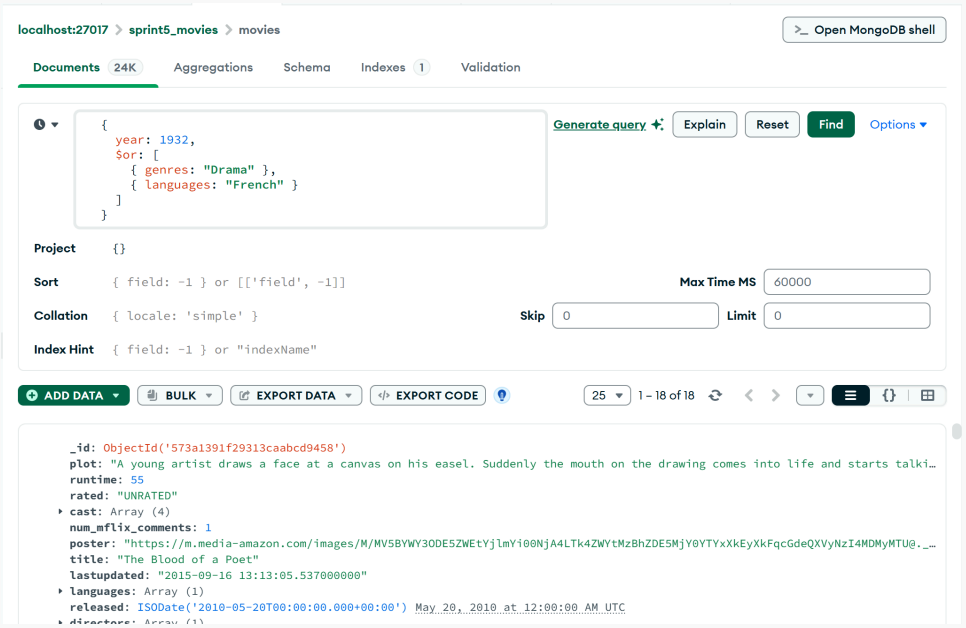
```
> db.movies.distinct("genres")
< [
    'Action',      'Adventure', 'Animation',
    'Biography',   'Comedy',    'Crime',
    'Documentary', 'Drama',     'Family',
    'Fantasy',     'Film-Noir', 'History',
    'Horror',      'Music',     'Musical',
    'Mystery',     'News',      'Romance',
    'Sci-Fi',      'Short',     'Sport',
    'Talk-Show',   'Thriller',  'War',
    'Western'
  ]
> db.movies.countDocuments({
    "genres":"Comedy"
  })
< 7024
```

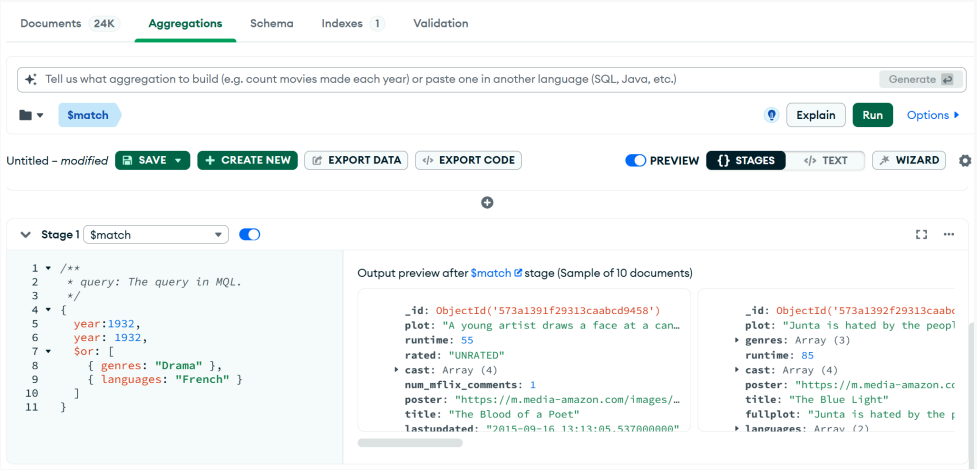




Exercise 2

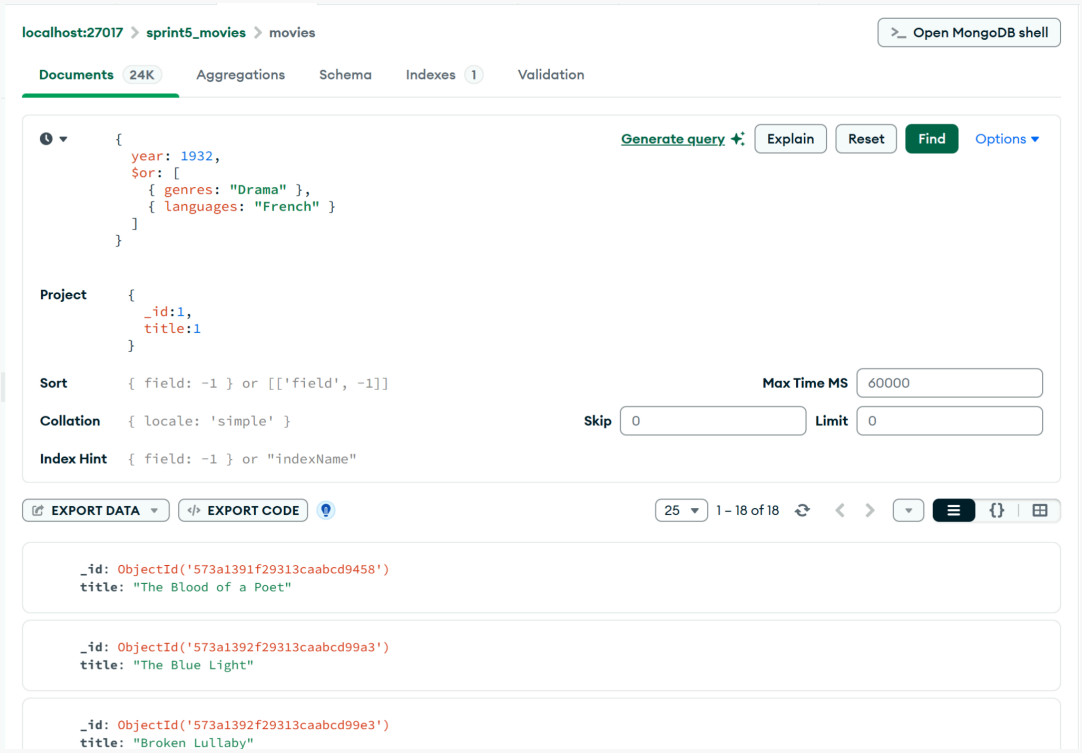
Show me all the documents for films produced in 1932, but the genre is drama or they are in French.






```
> db.movies.find({
  "year":1932,
  $or:[
    {"genres":"Drama"},
    {"languages":"French"}
  ]
 })
{
  _id: ObjectId('573a1391f29313caabcd9458'),
  plot: 'A young artist draws a face at a canvas on his easel. Suddenly the mouth on the drawing comes into life and starts t
  runtime: 55,
  rated: 'UNRATED',
  cast: [
    'Enrique Rivero',
    'Elizabeth Lee Miller',
    'Pauline Carton',
    'Odette Talazac'
  ],
```

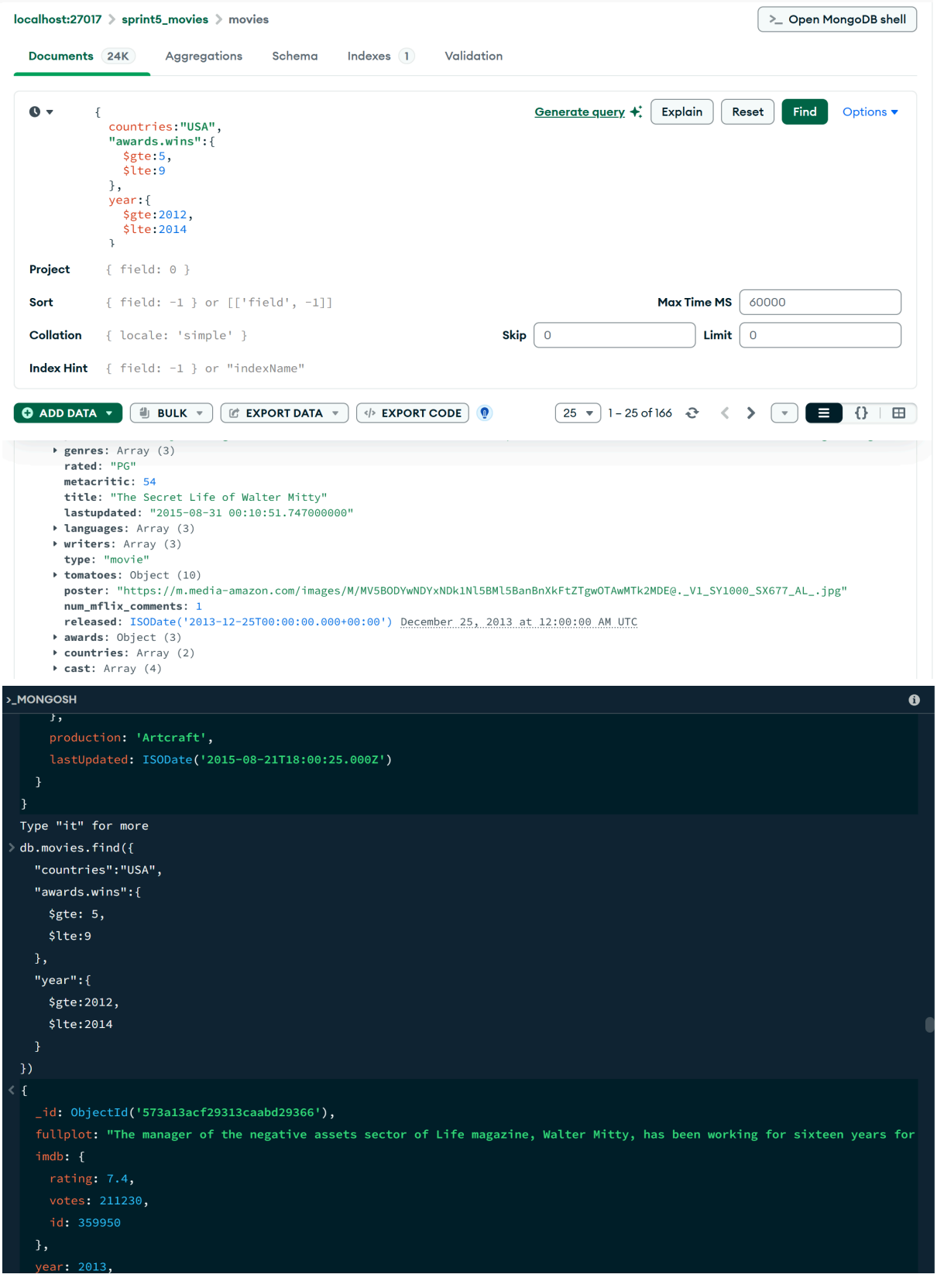
Only title and id of the film produced in 1932, but the genre is drama or they are in French.:

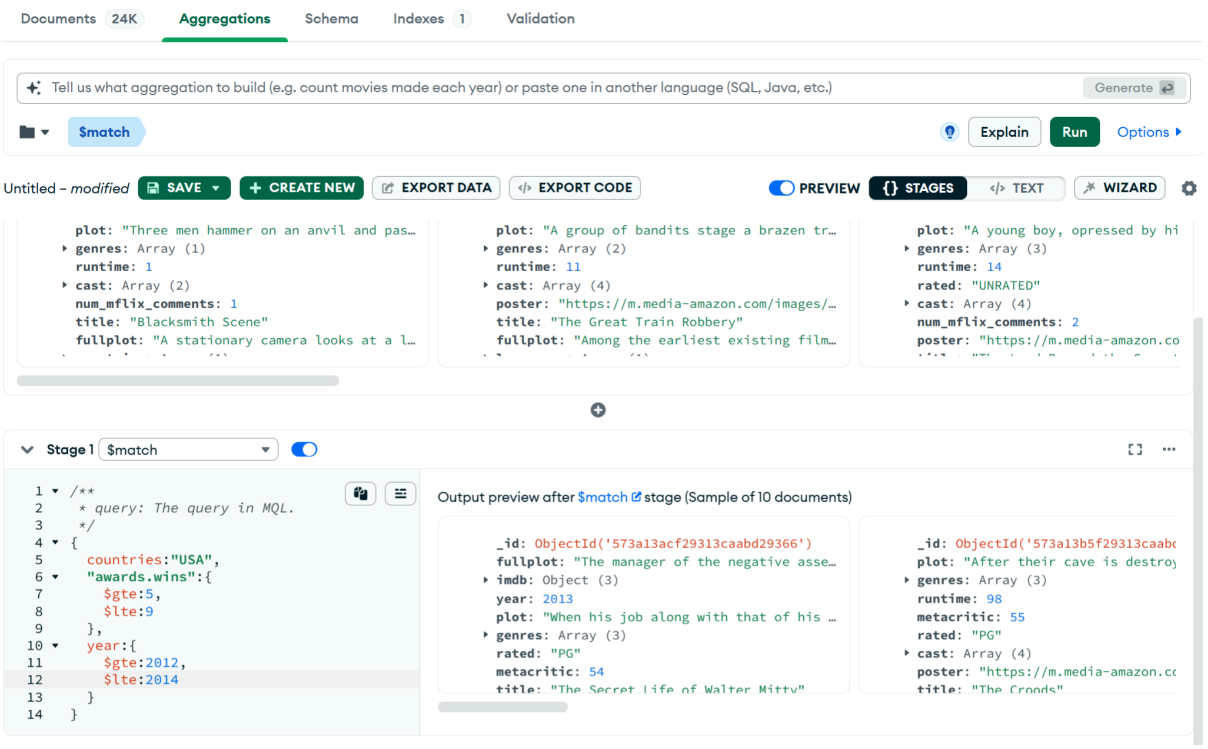


_id	title
573a1391f29313caabcd9458	The Blood of a Poet
573a1392f29313caabcd99a3	The Blue Light
573a1392f29313caabcd99e3	Broken Lullaby
573a1392f29313caabcd9a10	The Crowd Roars
573a1392f29313caabcd9a5d	A Farewell to Arms
573a1392f29313caabcd9a76	Forbidden
573a1392f29313caabcd9a7e	Freaks
573a1392f29313caabcd9aab	Grand Hotel
573a1392f29313caabcd9aff	I Am a Fugitive from a Chain Gang
573a1392f29313caabcd9be6	The Mummy
573a1392f29313caabcd9c1b	Payment Deferred
573a1392f29313caabcd9c4c	The Red Head
573a1392f29313caabcd9c5d	Red Dust
573a1392f29313caabcd9ca6	Scarface
573a1392f29313caabcd9caa	Shanghai Express
573a1392f29313caabcd9cce	Smilin' Through
573a1392f29313caabcd9d4a	Two Seconds
573a1392f29313caabcd9d4f	I Was Born, But...

Exercise 3

Show me all documents of American films with between 5 and 9 awards that were produced between 2012 and 2014.
166 films





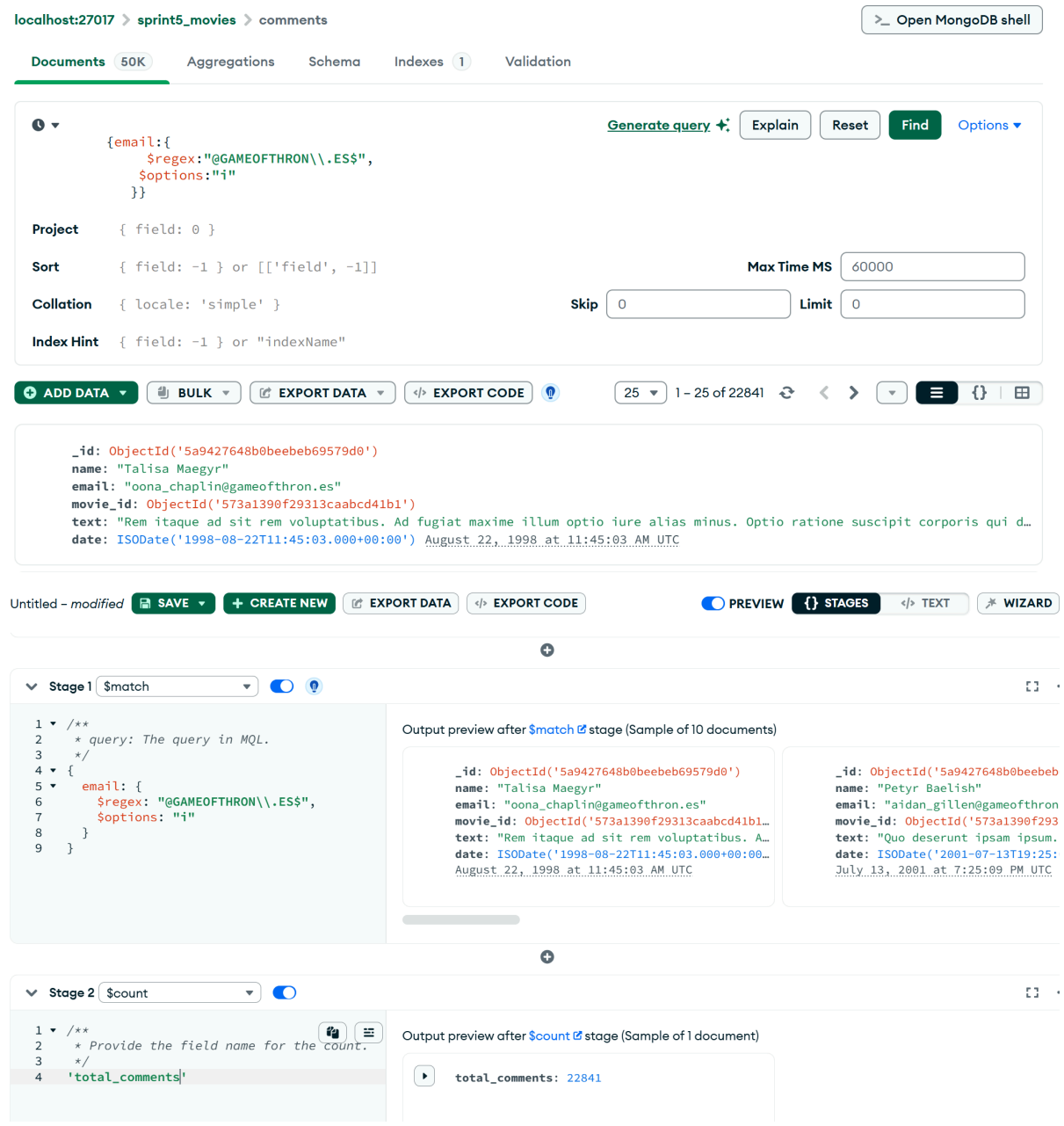
	A	B
1	_id	title
2	573a13acf29313caabd29366	The Secret Life of Walter Mitty
3	573a13b5f29313caabd45772	The Croods
4	573a13b9f29313caabd4dd15	The Book Thief
5	573a13b9f29313caabd4de39	World War Z
6	573a13b9f29313caabd4f21f	Godzilla
7	573a13bbf29313caabd5271d	Mr Hockey: The Gordie Howe Story
8	573a13bcf29313caabd57341	Hitchcock
9	573a13bef29313caabd5d9c2	Lone Survivor
10	573a13c1f29313caabd63af2	One Chance
11	573a13c1f29313caabd648d3	The Lone Ranger
12	573a13c1f29313caabd64b93	The World's End
13	573a13c1f29313caabd65df8	That's My Boy
14	573a13c1f29313caabd65e49	21 Jump Street
15	573a13c2f29313caabd66592	Mariachi Gringo
16	573a13c3f29313caabd6a560	Kill Your Darlings
17	573a13c3f29313caabd6b522	The Lucky One
18	573a13c5f29313caabd6fcf7	Particle Fever
19	573a13c5f29313caabd714a6	Star Trek Into Darkness
20	573a13c7f29313caabd742df	Prometheus
21	573a13c7f29313caabd7654a	Four Assassins
22	573a13c8f29313caabd778e9	The Paperboy
23	573a13c8f29313caabd78900	Heathens and Thieves
24	573a13c9f29313caabd7a37b	Line of Duty
25	573a13c9f29313caabd7ac64	The Mortal Instruments: City of Bones
26	573a13cbf29313caabd7f77a	To the Wonder
27	573a13cbf29313caabd81353	The Waiting Room
28	573a13ccf29313caabd81793	Cesar Chavez
29	573a13ccf29313caabd819c6	Oz the Great and Powerful
30	573a13cdf29313caabd85ad2	Stoker
31	573a13cef29313caabd869e0	Aquè Entre Nos
32	573a13cef29313caabd8735a	Battlestar Galactica: Blood & Chrome
33	573a13cef29313caabd880dd	The Discoverers
34	573a13cff29313caabd88ec2	The Way Way Back
35	573a13cff29313caabd8a55e	Found in Time
36	573a13d1f29313caabd8e193	This Is Martin Bonner
37	573a13d1f29313caabd8f943	Girl in Progress
38	573a13d2f29313caabd91592	The Truth About Emanuel
39	573a13d2f29313caabd917cc	Divergent

Level 2

Exercise 1

Count how many comments a user writes who uses "GAMEOFTHRON.ES" as their email domain. **22841**

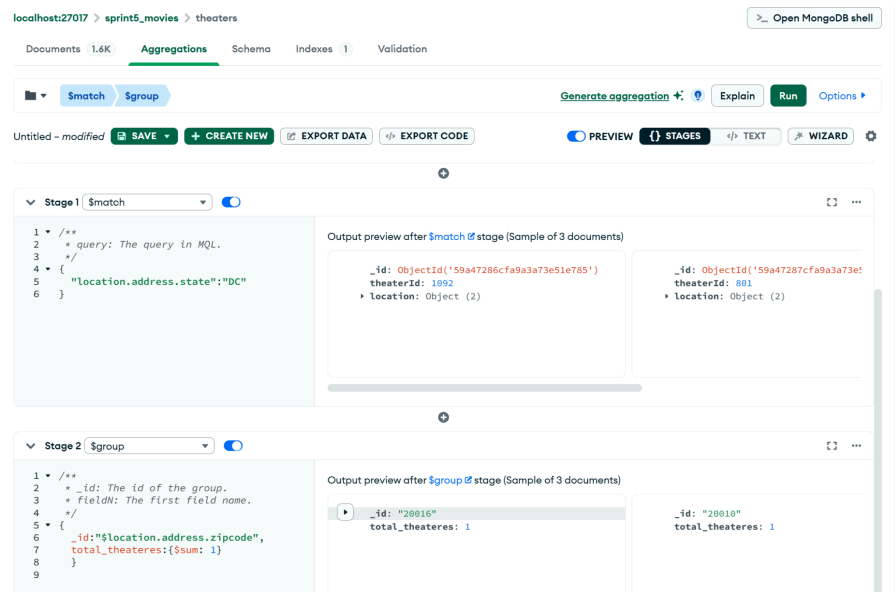
```
> db.comments.countDocuments({
    "email":{
      "$regex": "@GAMEOFTHRON\\.ES$",
              "$options": "i"
    }
  })
< 22841
sprint5_movies>
```

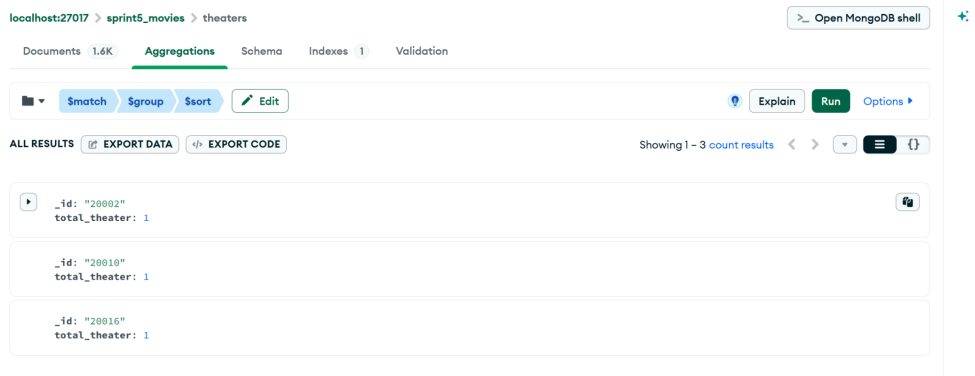


Exercise 2

How many movie theaters are there in each zip code located within the state of Washington DC (DC)?

```
>_MONGOSH
> db.theaters.aggregate([
    {
      $match: {
        "location.address.state": "DC"
      }
    },
    {
      $group: {
        _id: "$location.address.zipcode",
        total_theaters: { $sum: 1 }
      }
    },
    {
      $sort: {
        _id: 1
      }
    }
  ])
< {
    _id: '20002',
    total_theaters: 1
  }
  {
    _id: '20010',
    total_theaters: 1
  }
  {
    _id: '20016',
    total_theaters: 1
  }
sprint5_movies>
```

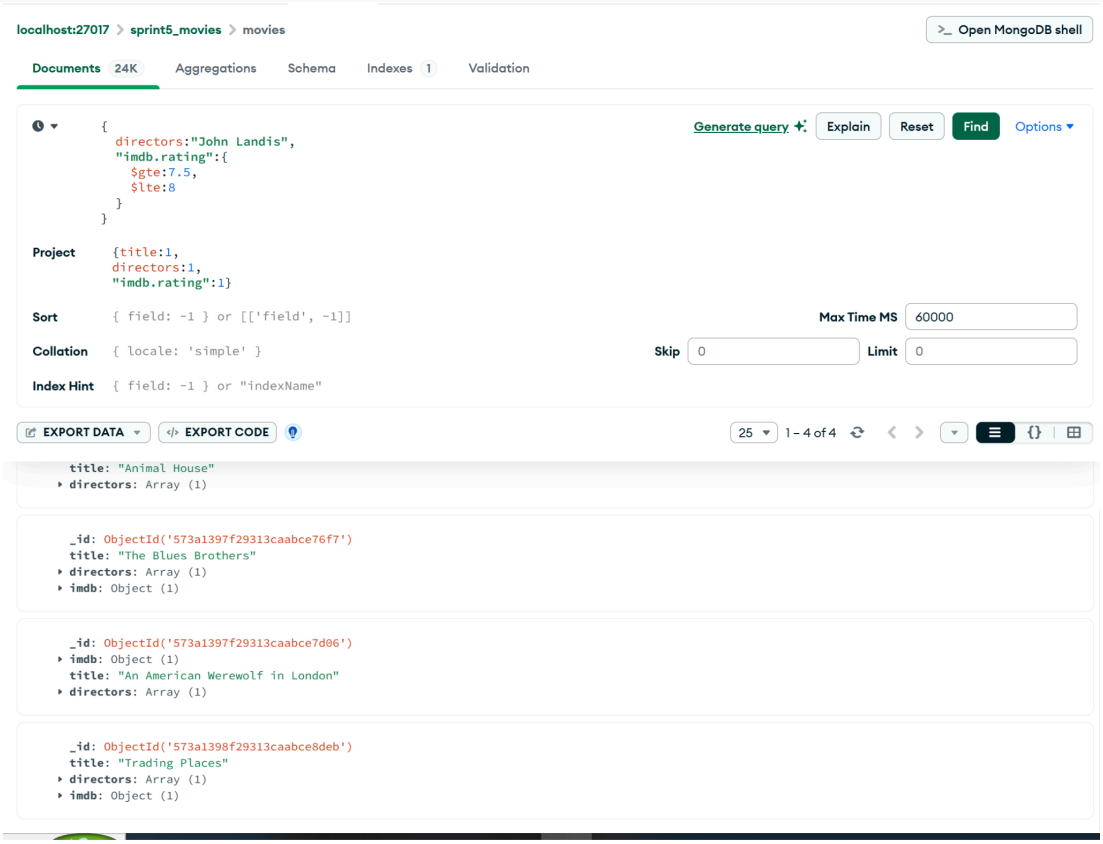


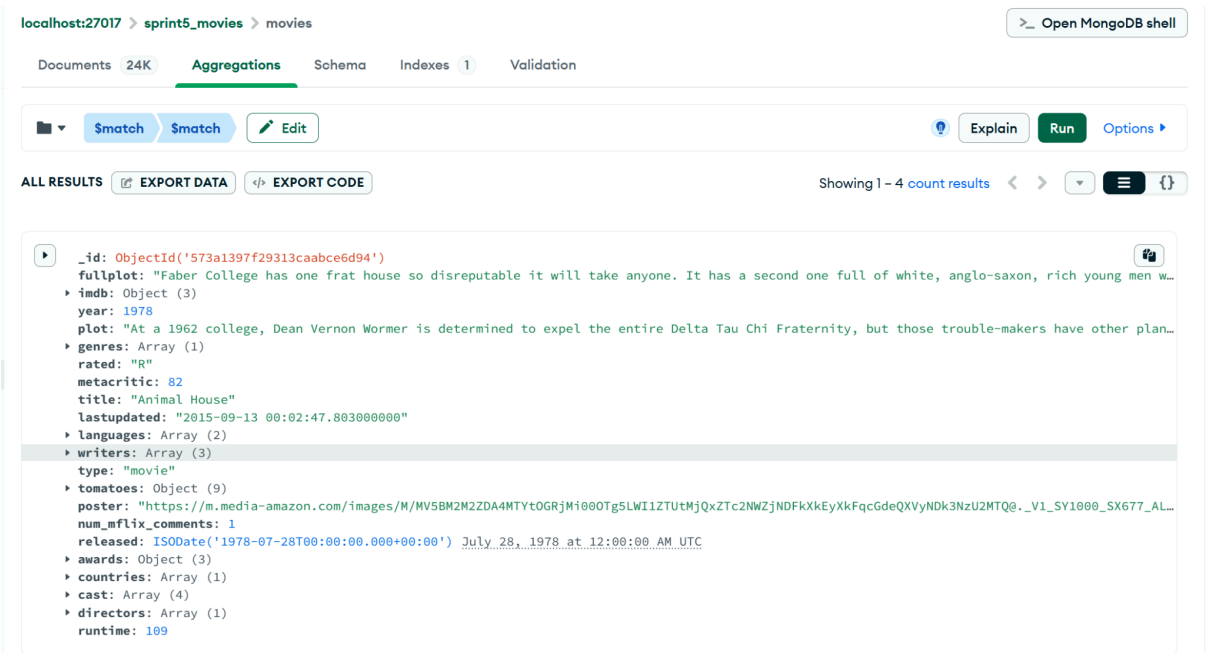


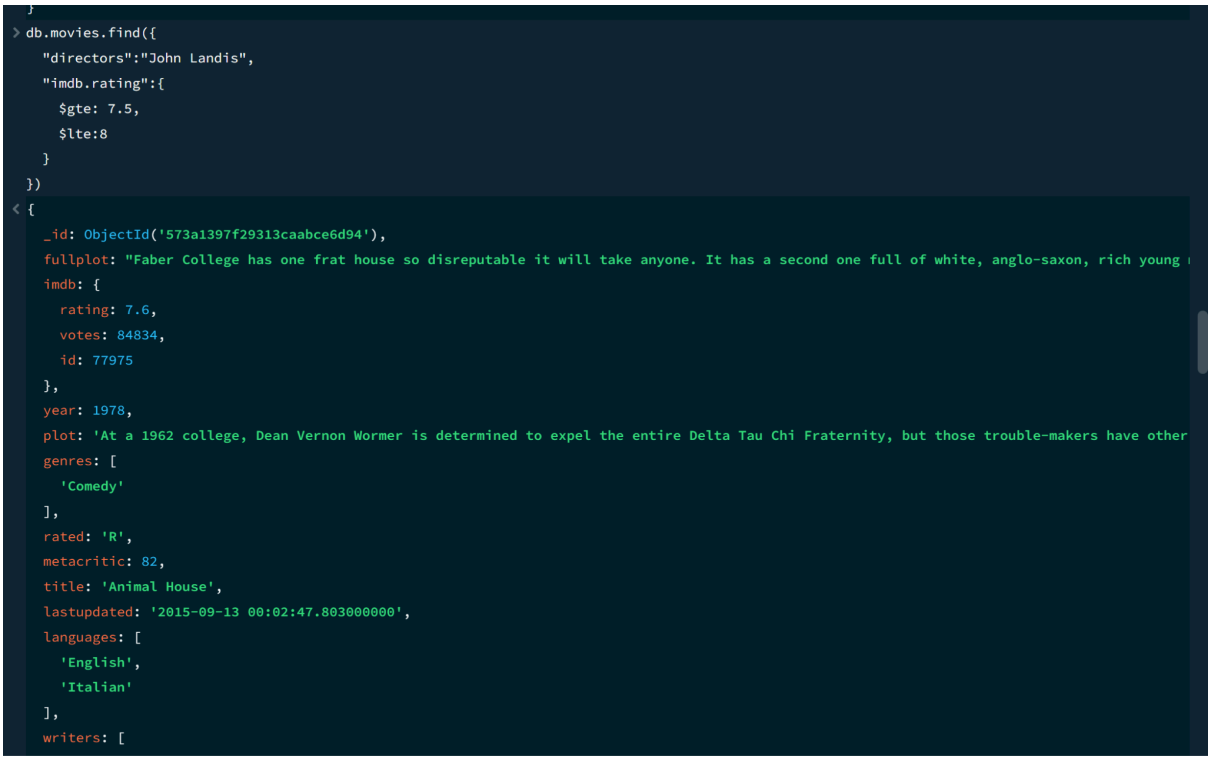
Level 3

Exercise 1

Find all films directed by John Landis with an IMDb (Internet Movie Database) rating between 7.5 and 8.






db.movies.find({
"directors":"John Landis",
"imdb.rating":{
$gte: 7.5,
$lte:8
}
})
{
_id: ObjectId('573a1397f29313caabce6d94'),
fullplot: "Faber College has one frat house so disreputable it will take anyone. It has a second one full of white, anglo-saxon, rich young
imdb: {
rating: 7.6,
votes: 84834,
id: 77975
},
year: 1978,
plot: 'At a 1962 college, Dean Vernon Wormer is determined to expel the entire Delta Tau Chi Fraternity, but those trouble-makers have other
genres: [
'Comedy'
],
rated: 'R',
metacritic: 82,
title: 'Animal House',
lastupdated: '2015-09-13 00:02:47.803000000',
languages: [
'English',
'Italian'
],
writers: [

Exercise 2

Shows the location of all the theaters in the database on a map.

